PREVALENCE OF HEPATITIS B VIRUS INFECTION, TOTAL ANTI-HBc POSITIVITY, AND ISOLATED ANTI-HBc AMONG 118,141 INDIVIDUALS UNDERGOING ROUTINE HEALTH CHECK-UPS AT MEDIC MEDICAL CENTER IN THE SOUTH OF VIETNAM

A/Prof. MD. PhD Thuy Thi Thu Pham[1*]; MD. Dat Tan Ho[1]; MD. Toan Bao Nguyen[1]; B.Sc. Vi Thi Tuong Nguyen[1]; MD. Hai Thanh Phan[1]

1. MEDIC Medical Center, Ho Chi Minh City, Viet Nam 2. Phan Chau Trinh University, Viet Nam

www.medic.com.vn

AASLD The Liver Meeting NOV. 5-9, 2026 Denver, CO

P 2226

BACKGROUND

Vietnam has one of the highest hepatitis B virus (HBV) infection burdens in the World Health Organization Western Pacific Region, with recent community HBsAg seroprevalence 7.5%, and HBV remains a leading etiology of hepatocellular carcinoma (HCC). In 2023, the U.S. Centers for Disease Control and Prevention (CDC) recommended universal adult HBV screening using a triple serologic panel (HBsAg, anti-HBs, and total anti-HBc). However, evidence on the revalence and clinical implications of isolated anti-HBc positivity (iAHBc) - particularly its associations with HCC and liver fibrosis - remains scarce in the Vietnamese population.

OBJECTIVES:

(1). Determine the prevalence of HBV serologic subgroups identified by the triple test, particularly iAHBc, among individuals in Southern Vietnam;

(2). Assess the associations of HBsAg positivity, anti-HBc positivity, and iAHBc with age and sex;

(3). Examine the associations between HBV serologic subgroups and both liver fibrosis severity and HCC risk, thereby providing evidence to inform the potential inclusion of anti-HBc in HBV screening programs in Vietnam.

METHODS AND MATERIALS

- In this cross-sectional study, individuals attending routine health check-ups at MEDIC Medical Center, Ho Chi Minh City (January 2023-December 2025), were simultaneously tested for HBsAg, anti-HBs, and anti-HBc; those with hepatitis C virus infection were excluded.
- Participants were classified into five serologic groups: at risk (all negative), vaccine-induced immunity (anti-HBs only), resolved infection (anti-HBs and anti-HBc positive, HBsAg negative), iAHBc (anti-HBc only), and current infection (HBsAg positive).
- Liver fibrosis was staged by FibroScan (F0-F4), and HCC was ascertained from clinical records.
- Distributions across sex, age, and serologic subgroups were compared using chi-square tests.
- Multivariable logistic regression estimated prevalence odds ratios (PORs) and 95% confidence intervals (CIs) for associations of serologic subgroups with age, sex, liver fibrosis, and HCC.
- Significance level was set at $\alpha = 0.05$, with analyses performed in R version 4.6.0.

RESULTS

Table 1: Characteristics of 118,141 subjects in the study

	Overall N = 118,141	Female N = 65,573	Male N = 52,568	p-value*
	100%	55.5%	44.5%	
Age (mean, SD)	44 ± 14	45 ± 14	43 ± 14	< 0.001
< 20	3,689 (3.1%)	1,752 (2.7%)	1,937 (3.7%)	< 0.001
20 - 29	15,962 (13.5%)	8,749 (13.3%)	7,213 (13.7%)	
30 - 39	26,162 (22.1%)	13,746 (21.0%)	12,416 (23.6%)	
40 - 49	30,041 (25.4%)	16,459 (25.1%)	13,582 (25.8%)	
50 - 59	25,212 (21.3%)	14,521 (22.1%)	10,691 (20.3%)	
60+	17,075 (14.5%)	10,346 (15.8%)	6,729 (12.8%)	
HBsAg				< 0.001
HBsAg (-)	110,751 (93.7%)	62,116 (94.7%)	48,637 (92.5%)	
HBsAg (+)	7,388 (6.3%)	3,457 (5.3%)	3,931 (7.5%)	
Anti HBs				< 0.001
Anti - HBs (-)	51,869 (43.9%)	28,359 (43.2%)	23,510 (44.7%)	
Anti - HBs (+)	66,272 (56.1%)	37,214 (56.8%)	29,058 (55.3%)	
Anti - HBc total				< 0.001
Anti - HBc (-)	65,985 (55.9%)	38,134 (58.2%)	27,851 (53.0%)	
Anti - HBc (+)	52,156 (44.1%)	27,439 (41.8%)	24,717 (47.0%)	
HCC diagnosis	179 (0.15%)	49 (0.07%)	130 (0.25%)	< 0.001
Fibro Scan				< 0.001
F0 - F1	15,818 (77.3%)	6,590 (77.6%)	9,228 (77.0%)	
F2	2,279 (11.1%)	883 (10.4%)	1,396 (11.7%)	
F3	1,377 (6.7%)	621 (7.3%)	756 (6.3%)	
F4	993 (4.9%)	396 (4.7%)	597 (5.0%)	
Unknown	97,674	57,083	40,591	

Table 2: HBV infection status based on triple panel testing, stratifying by gender

	Triple panel	Total	Female	Male	p-value*
		N = 118141	N = 65573	N = 52568	
At risk	HBsAg (-) Anti - HBs (-) Anti - HBc (-)	37288 (31.6%)	21474 (32.7%)	15814 (30.1%)	< 0.0001
Vaccine - induced immunity	HBsAg (-) Anti - HBs (+) Anti - HBc (-)	28697 (24.3%)	16660 (25.4%)	12037 (22.9%)	< 0.0001
Resolved HBV infection	HBsAg (-) Anti - HBs (+) Anti - HBc (+)	37412 (31.7%)	20481 (31.2%)	16931 (32.2%)	< 0.0001
iAHBc	HBsAg (-) Anti - HBs (-) Anti - HBc (+)	7356 (6.2%)	3501 (5.3%)	3855 (7.3%)	< 0.0001
Total HBsAg (+)	HBsAg (+)	7388 (6.3%)	3457 (5.3%)	3931 (7.5%)	< 0.0001

Table 3: HBV infection status based on triple panel testing, stratifying by age groups

	Triple panel	<20 (N = 3689)	20 - 29 (N = 15962)	30 - 39 (N= 26162)	40 - 49 (N = 30041)	50 - 59 (N = 25212)	60+ (N = 17075)	p - value*
Age (years, SD)		16.90 ± 2.64	24.89 ± 2.82	34.71 ± 2.82	44.46 ± 2.86	54.21 ± 2.80	66.16 ± 5.46	<0.0001
Male (yes)		1937 (52.5%)	7213 (45.2%)	12416 (47.5%)	13582 (45.2%)	10691 (42.4%)	6729 (39.4%)	<0.0001
At risk	HBsAg (-) Anti - HBs (-) Anti - HBc (-)	2144 (58.1%)	8255 (51.7%)	10279 (39.3%)	8558 (28.5%)	5246 (20.8%)	2806 (16.4%)	<0.0001
Vaccine - induced immunity	HBsAg (-) Anti - HBs (+) Anti - HBc (-)	1369 (37.1%)	5644 (35.4%)	7140 (28.1%)	6906 (23.0%)	4787 (19.0%)	2651 (15.5%)	<0.0001
Resolved HBV infection	HBsAg (-) Anti - HBs (+) Anti - HBc (+)	68 (1.8%)	1301 (8.2%)	6248 (23.9%)	10650 (35.5%)	11103 (44.0%)	8040 (47.1%)	<0.0001
iAHBc	HBsAg (-) Anti - HBs (-) Anti - HBc (+)	20 (0.5%)	157 (1.0%)	673 (2.6%)	1679 (5.6%)	2309 (9.2%)	2518 (14.7%)	<0.0001
Current HBV infection	HBsAg (+)	88 (2.4%)	603 (3.8%)	1622 (6.2%)	2248 (7.5%)	1767 (7.0%)	1060 (6.2%)	<0.0001

Table 4: Comparison of the distribution of iAHBc and at-risk populations by FibroScan report and HCC

	FibroScan (N = 20,467)			p - value*
	F0 - F1	F2	F3 - F4	
At - risk (N = 4411)	3612 (81.85%)	390 (8.84%)	411 (9.31%)	< 0.001
iAHBc (N = 2011)	1270 (63.15%)	300 (14.92%)	441 (21.93%)	
	HCC (N = 118,141)			
	Non - HCC	HCC		
At - risk (N = 37,288)	37269 (99.95%)	19 (0.05%)		< 0.001
iAHBc (N = 7356)	7317 (99.47%)	39 (0.53%)		

*Groups were compared using Chi-squared tests. For FibroScan, only those with a FibroScan report (n=20,467) were included in the analysis. *P*-value <0.05 denotes significant difference.

Figure 1: Prevalence of different serological patterns across age subgroups

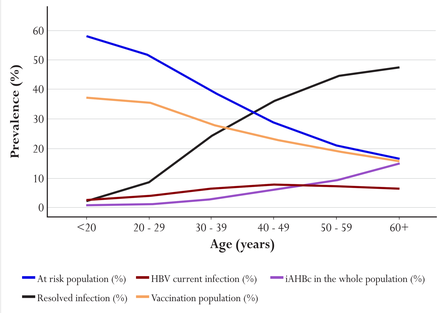

Table 5: Association of HBsAg positivity, Anti-HBc positivity, and iAHBc with age and sex

	HBsAg (+)		Anti - HBc (+)		iAHBc	
Age	POR (95% CI)	p - value*	POR (95% CI)	p - value*	POR (95% CI)	p - value*
<20	1.00		1.00		1.00	
20 - 29	1.65 (1.33 - 2.09)	< 0.001	3.04 (2.60 - 3.58)	<0.001	1.88 (1.21 - 3.09)	0.007
30 - 39	2.76 (2.24 - 3.46)	< 0.001	9.92 (8.53 - 11.60)	<0.001	4.96 (3.26 - 8.00)	<0.001
40 - 49	3.41 (2.77 - 4.26)	< 0.001	19.51 (16.8 - 22.82)	<0.001	11.24 (7.44 - 18.09)	<0.001
50 - 59	3.21 (2.60 - 4.02)	< 0.001	31.71 (27.28 - 37.10)	<0.001	19.41 (12.85 - 31.24)	<0.001
60+	2.85 (2.30 - 3.58)	< 0.001	45.16 (38.80 - 52.91)	<0.001	31.85 (22.41 - 54.47)	<0.001
Sex						
Female	1.00	< 0.001	1.00	<0.001	1.00	<0.001
Male	1.47 (1.40 - 1.54)		1.39 (1.36 - 1.43)		1.55 (1.48 - 1.63)	

*Logistic regression model adjusted for sex and age subgroups. POR: prevalence odds ratio. *P*-value <0.05 denotes significant difference.

Table 6: Association between different serology status and risk of having HCC, stratified by sex

	Total#	Male	Female
	POR (95% CI)	POR (95% CI)	POR (95% CI)
At risk	1.00	1.00	1.00
Resolved HBV infection	0.97 (0.57 - 1.71)	0.99 (0.51 - 2.09)	0.98 (0.41 - 2.47)
iAHBc	2.77 (1.59 - 4.98) **	2.85 (1.41 - 6.05) *	2.90 (1.11 - 7.84) *
Current HBV infection	9.91 (6.05 - 17.0) **	11.62 (6.29 - 23.53) **	7.17 (3.05 - 18.05) **
Vaccine - induced immunity	0.48 (0.20 - 1.06)	0.52 (0.16 - 1.43)	0.43 (0.09 - 1.48)

Logistic regression model adjusted for sex and age groups. POR: prevalence odds ratio. *p*-value <0.01, **p-value <0.001. *P*-value <0.05 denotes significant difference.

CONCLUSIONS

To our knowledge, this is the largest study in Vietnam to apply a triple HBV serologic testing panel. The prevalence of current HBV infection in Southern Vietnam was 6.3%, and 6.2% of participants had iAHBc - a subgroup that would likely be missed by the conventional two-marker screening approach. Importantly, iAHBc was associated with 2.77-fold higher odds of HCC and a substantially higher prevalence of advanced fibrosis (F3–F4) compared with triple - seronegative individuals, indicating that this serologic profile is not clinically benign. These findings support consideration of the triple-marker panel for routine HBV screening in Vietnam.

References

1. C. Freeland et al. The importance of triple panel testing for hepatitis B and the burden of isolated anti - hepatitis B core antibodies within a community sample. Journal of Virus Eradication 9 (2023) 100358.
2. Coppola et al. Association between anti - HBc positivity and hepatocellular carcinoma in HBsAg - negative subjects with chronic liver disease Medicine (2016) 95:30.
3. Olga V. Kalinina et al. Parenteral viral hepatitis among students in Da Nang, Vietnam, and insights into new hepatitis B virus genotype B classification. Scientific Reports | (2025) 15:21589.
4. Shuai - Wen Huang et al. Prevalence of Cirrhosis/Advanced Fibrosis Among HBsAg - Negative and HBcAb - Positive US Adults: A Nationwide Population-Based Study. Infect Dis Ther (2022) 11:1901–1916.
5. Trang N. D. P. et al. Establishing baseline framework for hepatitis B virus micro - elimination in Ho Chi Minh City, Vietnam - A community - based seroprevalence study. www.thelancet.com Vol 30 January, 2023.
6. Won Sohn et al. Isolated Hepatitis B Core Antibody Positivity and Long - Term Liver - Related Mortality in Korea: A Cohort Study. American Journal of Gastroenterology 118(1):p 95-104, January 2023.

Contact Information: drthuthuy@gmail.com